Trainings

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Note : If an air compressor is **not** mounted on the engine, it is **not** necessary to drain the cooling system.

Note : Some accessory drives may differ slightly from the one highlighted in the procedure below for example couplings and mounting hardware may differ.

- Drain the cooling system, if applicable. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Remove the fuel pump. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html)
- Remove the air compressor, if applicable. Refer to Procedure 012-014 in Section 12. (/qs3/pubsys2/xml/en/procedures/56/56-012-014-tr.html)
- Remove the accessory drive pulley, if applicable. Refer to Procedure 009-004 in Section 9. (/qs3/pubsys2/xml/en/procedures/56/56-009-004-tr.html)
- Remove the accessory drive cover, if applicable. Refer to Procedure 009-039 in Section 9.

(/qs3/pubsys2/xml/en/procedures/56/56-009-039-tr.html)

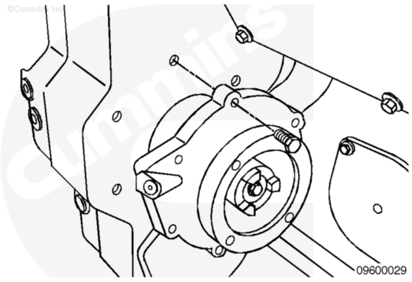
- Remove the accessory drive seal, if applicable. Refer to Procedure 001-003 in Section 3. (/qs3/pubsys2/xml/en/procedures/56/56-001-003-tr.html)

Remove

Remove the six capscrews.

Remove the accessory drive assembly.

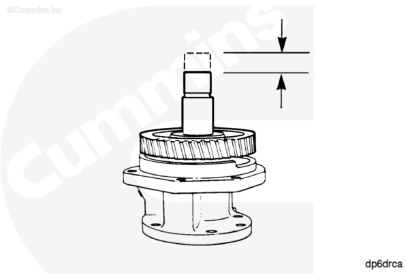
Remove and inspect the gasket.



Initial Check

Use a dial indicator to measure the accessory drive shaft end clearance.

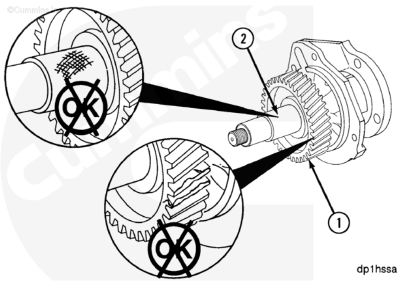
Accessory Shaft End Clearance		
mm		in
0.05	MIN	0.002
0.30	MAX	0.0118



Inspect the housing for cracks or damaged mounting holes.

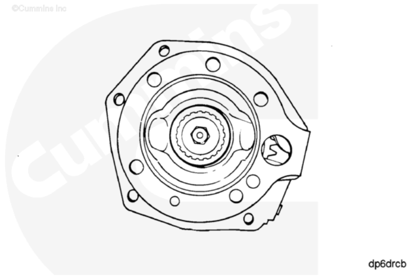
Inspect the drive gear (1) for damaged teeth.

Inspect the shaft (2) for scratches, scores, or other damage



Inspect the coupling washer.

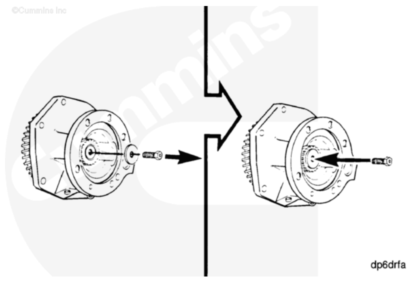
Note : The coupling washer **must** be positioned tightly between the coupling and the shaft. If the washer is **not** tight, the drive unit **must** be rebuilt or replaced.



Disassemble

⚠ CAUTION ⚠

To prevent damage to the shaft, the capscrew must be installed into the drive shaft without the washer.

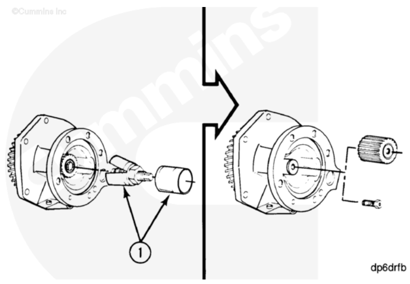


Remove the capscrew and washer.

Install the capscrew into the shaft.

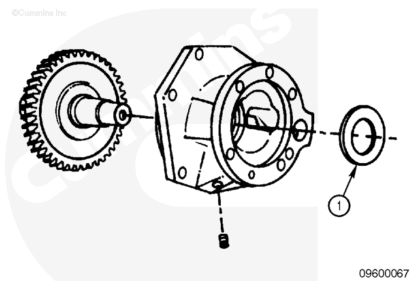
Use a jaw puller to remove the type coupling.

Remove the capscrew from the shaft.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Remove the clamping washer (1).

Remove the gear and shaft assembly.

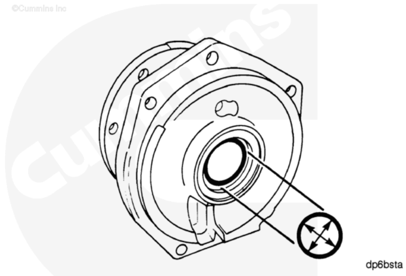
Remove the pipe plug from the housing.

Use solvent to clean the parts.

Measure the inside diameter of the bushings.

Accessory Drive Bushing Inside Diameter		
mm		in
49.95	MIN	1.9272
50.03	MAX	1.9697

If the inside diameter of either bushing is **not** within specification, both bushings **must** be replaced.

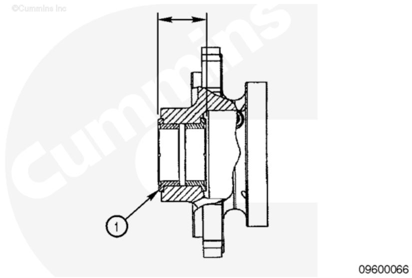


Check the grooved surfaces of the bushings (1) for damage.

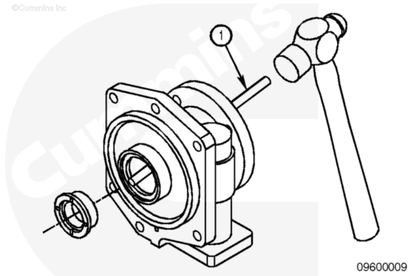
Measure the distance from the outside of the thrust faces of the two bushings.

Accessory Drive Busing Thrust Face Distance		
mm		in
45.370	MIN	1.7862
45.450	MAX	1.7894

If the distance between the two thrust faces is **not** within specification, both bushings **must** be replaced.

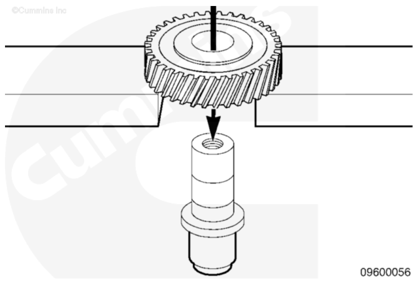


Remove the flanged bushing from the drive housing using a brass drift (1), taking care **not** to damage the bushing bore.



Only remove the gear from the shaft when the gear or the shaft **must** be replaced.

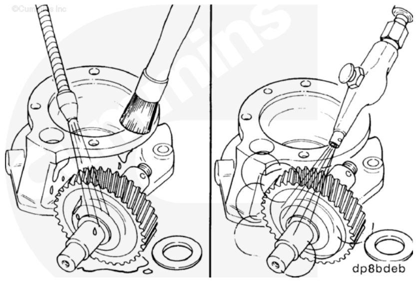
Use an arbor press to remove the gear.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



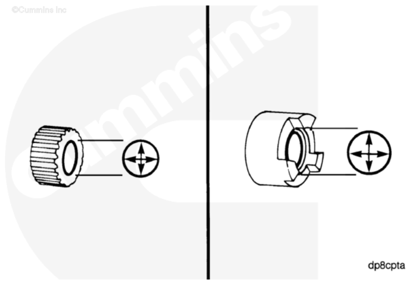
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

Use solvent to clean the parts.

Dry the parts with compressed air.

Check the inner diameter of the coupling for damage.



If damage is found, the coupling **must** be replaced.

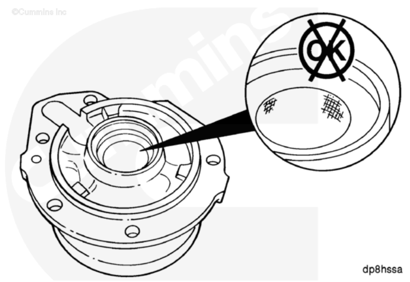
Measure the inside diameter of the coupling.

Accessory Drive Lovejoy Coupling Inside Diameter		
mm		in
25.377	MIN	0.9991
25.407	MAX	1.0003

Accessory Drive Spline Coupling Inside Diameter		
mm		in
25.255	MIN	0.9943
25.278	MAX	0.9952

If the inner diameter of the coupling is **not** within specification, the coupling **must** be replaced.

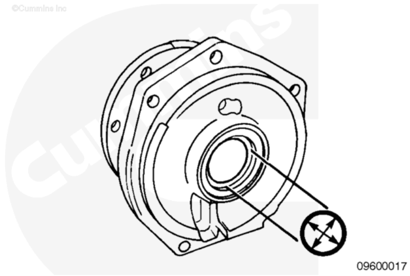
If the bushing was removed, inspect the housing for scoring and other damage.



Replace the housing if damaged.

If the bushing was removed, measure the inside diameter of the bushing bore.

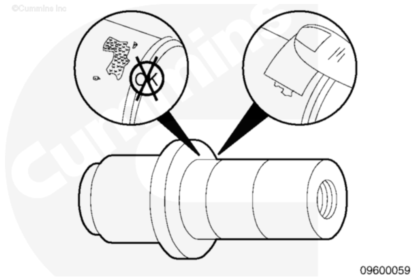
Accessory Drive Bushing Bore Inside Diameter		
mm		in
60.00	MIN	2.3622
60.05	MAX	2.3642



Replace the housing if the bushing bore is **not** within specifications.

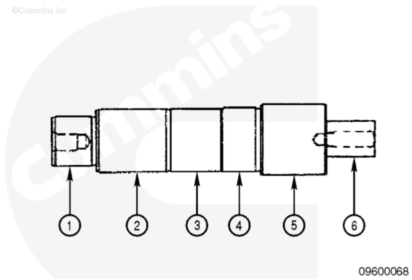
Inspect the shaft in the gear fit area for fretting or burrs.

Clean burrs with emery cloth.



Measure the shaft outside diameter.

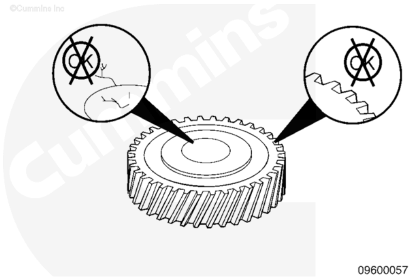
Accessory Drive Shaft Outside Diameter Measurements			
	mm		in
(1)	34.964	MIN	1.3765
	34.976	MAX	1.3770
(2)	43.928	MIN	1.7295
	43.940	MAX	1.7299
(3)	44.370	MIN	1.7469
	44.630	MAX	1.7571
(4)	45.070	MIN	1.7744
	45.082	MAX	1.7749
(5)	49.820	MIN	1.9614
	49.832	MAX	1.9619
(6)	25.424	MIN	1.0019
	25.436	MAX	1.0014



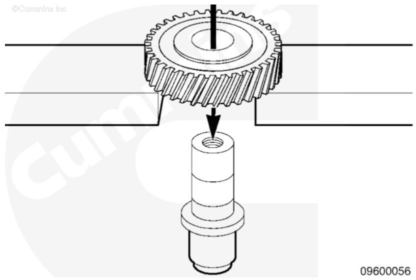
If the outside diameter is **not** within specification, the shaft **must** be replaced.

Check the gear teeth and gear bore for damage.

Replace the gear if damaged.



Measure the gear inside diameter.



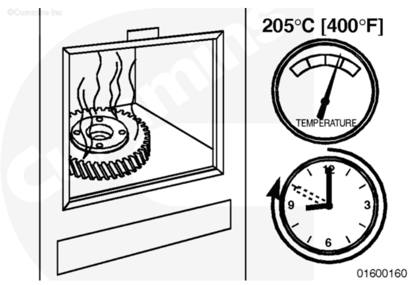
Accessory Drive Gear Inside Diameter		
mm		in
44.98	MIN	1.7709
45.02	MAX	1.7724

If the gear is **not** within specification, the gear **must** be replaced.

Assemble

⚠ WARNING ⚠

To reduce the possibility of personal injury, wear protective gloves when handling parts that have been heated.



⚠ CAUTION ⚠

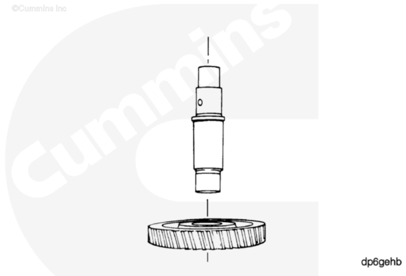
Do not exceed the specified time or temperature. Damage to the gear teeth will result.

If an adequate press is **not** available, an oven can be used.

Use an oven with the temperature adjusted to 205°C [400°F]. Heat the gear in the oven for a minimum of 40 minutes but **not** more than a maximum of one hour. The inner diameter of the gear will become larger and will simplify the installation of the gear on the camshaft.

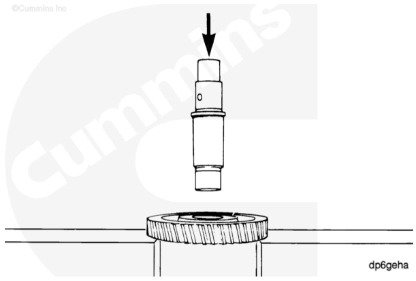
⚠ WARNING ⚠

To reduce the possibility of personal injury, wear protective gloves when handling parts that have been heated.



Support the gear.

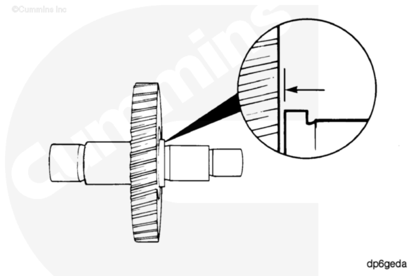
Slide the shaft into the gear until the shoulder of the shaft touches the gear.



If an arbor press is available, support the gear.

Lubricate the shaft with clean engine oil.

Press the shaft through the gear until the shoulder of the shaft touches the gear.



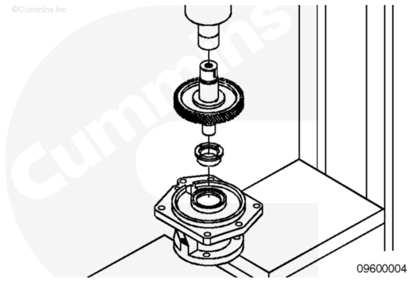
⚠ CAUTION ⚠

If the heat method was used to install the gear, allow the gear to air cool. Do not use water or oil to reduce cooling time. Damage to the gear can result.

Use a feeler gauge to measure the distance between the shoulder of the shaft and the gear.

Accessory Drive Gear to Shaft Distance		
mm		in
0.05	MAX	0.002

If the distance between the gear and the shaft is **not** within specification, press the gear on until the specification is met.



⚠ CAUTION ⚠

The bushings must be pressed in square to the bores of the housing. The thrust faces of the two bushings must be parallel and concentric. The bushing with the center counterbored must have the pressure applied to the counterbore face and not on the thrust bearing face. Failure to comply can result in distortion of the thrust face.

Support the housing.

The bushing with the smaller outside diameter is pressed into the output side.

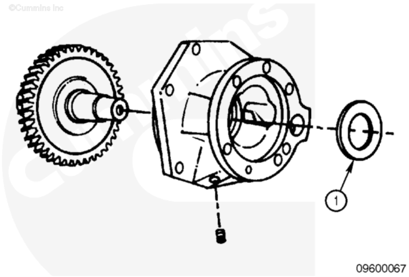
Use an arbor press and the shaft as a mandrel to press the bushing into the housing.

Turn the housing over and press the remaining bushing into place.

Lubricate the grooved surface of the thrust bearing with Lubriplate™ 105 or equivalent.

Install the gear and shaft assembly into the housing.

Install the clamping washer (1) with the beveled edge up.



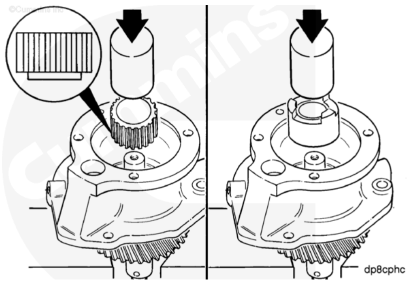
Support the assembly on the gear or the shaft.

Use an arbor press and a mandrel to press the coupling onto the shaft until it touches the clamping washer.

The clamping washer **must** be positioned tightly between the coupling and the shoulder of the shaft.

Install the pipe plug into the housing.

Tighten the pipe plug.

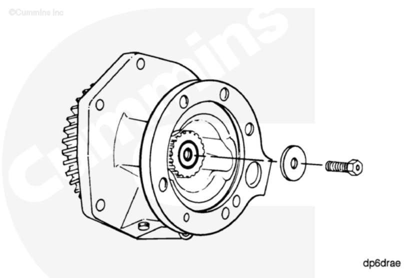


Torque Value: 8 n•m [75 in-lb]

The capscrew can contain an oil drilling if an air compressor is to be mounted on the engine.

Install the washer and capscrew.

Tighten the capscrew.

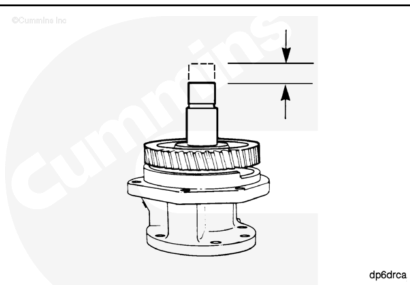


Torque Value:
9.525 mm [0.375 in] Capscrew Length 45 n•m [35 ft-lb]

Torque Value:
12.7 mm [0.50 in] Capscrew Length 100 n•m [75 ft-lb]

Rotate the shaft, checking for correct assembly.

Use a dial indicator to measure the accessory drive shaft end clearance.



Accessory Drive Shaft End Clearance		
mm		in
0.05	MIN	0.002
0.30	MAX	0.0118

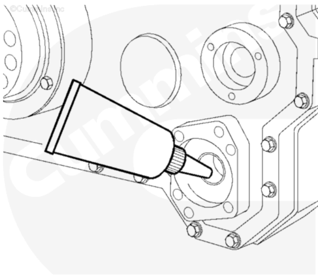
If the end clearance is **not** within specifications, make sure the coupling is positioned tightly against the clamping washer.

Install

with Mechanically Actuated Injector

⚠ CAUTION ⚠

Do not use sealants on the gasket. The gasket is manufactured from material that sealants will damage.

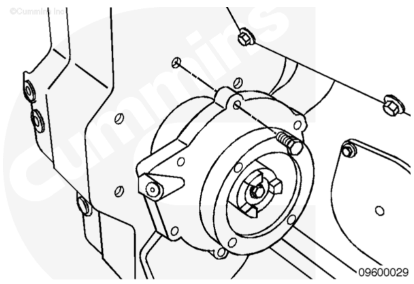


09600037

Lubricate the bushing in the front gear housing cover with Lubriplate™ 105.

Install the gasket onto the pilot of the drive housing cover.

There is **not** a requirement to align the index on the drive gear with the marks on the camshaft gear on QSK45 and QSK60 engines.



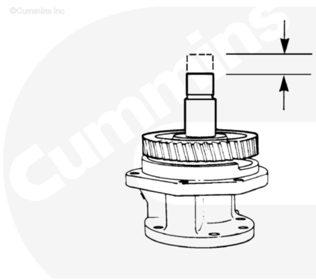
09600029

Install the accessory drive assembly, turning the shaft as necessary to engage the gears.

Install and tighten the six capscrews.

Torque Value: 45 n•m [33 ft-lb]

With the accessory drive installed, measure the end clearance.



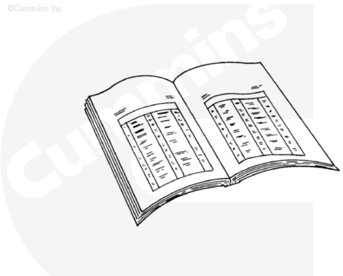
dp6drca

Accessory Drive End Clearance		
mm		in
0.13	MIN	0.0051
0.29	MAX	0.0114

Finishing Steps

⚠ CAUTION ⚠

When installing the air compressor, the use of the wrong spline coupling will cause damage to the accessory driveshaft.



- Install the accessory drive seal, if applicable. Refer to Procedure 001-003 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-003-tr.html)
- Install the accessory drive pulley, if applicable. Refer to Procedure 009-004 in Section 9. (/qs3/pubsys2/xml/en/procedures/56/56-009-004-tr.html)
- Install the accessory drive cover, if applicable. Refer to Procedure 009-039 in Section 9. (/qs3/pubsys2/xml/en/procedures/56/56-009-039-tr.html)
- Install the air compressor, if applicable. Refer to Procedure 012-014 in Section 12. (/qs3/pubsys2/xml/en/procedures/56/56-012-014-tr.html)
- Install the fuel pump. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html)
- Fill the cooling system, if applicable. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Start and operate the engine until the coolant temperature reaches 71°C [160°F]. Check for coolant leaks.

Last Modified: 11-Oct-2016